

 **OFFICIAL DOCUMENTATION**

Smart Household Energy User Manual

Learn how to analyze electricity bills, monitor energy usage, discover savings, and optimize your home's energy consumption using AI-powered insights.

30%

Avg. Energy Savings

98%

AI Accuracy

50k+

Homes Optimized

4.9★

User Rating

 DOCUMENTATION

Quick Start Guides

Jump to the section you need, or follow the complete onboarding flow below.



Dashboard Overview

Understand your energy score, live stats, and quick action buttons from the central command center.

[Explore →](#)

Bill Analyzer

Upload and scan your electricity bill with AI-powered OCR and Gemini Vision recognition.

[Explore →](#)



AI Recommendations

Get personalized energy-saving tips ranked by impact and potential monthly savings.

[Explore →](#)



Solar ROI Calculator

Model your solar investment payback period and 25-year cumulative savings projection.

[Explore →](#)

ONBOARDING

Getting Started in 5 Steps

From account creation to your first AI-powered energy report — follow these steps to get up and running.

1

Create Your Account

Visit **app.smartenergyai.com** and sign up with your email address or Google account. Verify your email to activate your profile.



Account Creation Screen

2

Complete Your Profile

Enter your household details: home size, number of occupants, location, and your electricity provider (DISCOM). This helps the AI calibrate accurate predictions.



Pro Tip

A complete profile improves AI recommendation accuracy by up to **40%**. Don't skip the appliance list!

3

Upload Your Electricity Bill

Go to the **Bill Analyzer** tab and upload your latest electricity bill as a PDF, JPG, or PNG. Our AI will extract key information automatically.



Bill Upload Interface

4

Run the AI Energy Audit

Navigate to **Home Energy Audit** and list your household appliances with daily usage hours to generate a real-time consumption breakdown.



Energy Audit Interface

View AI Recommendations

Check the **AI Recommendations** tab for personalized energy-saving tips, ranked by potential savings and confidence score.



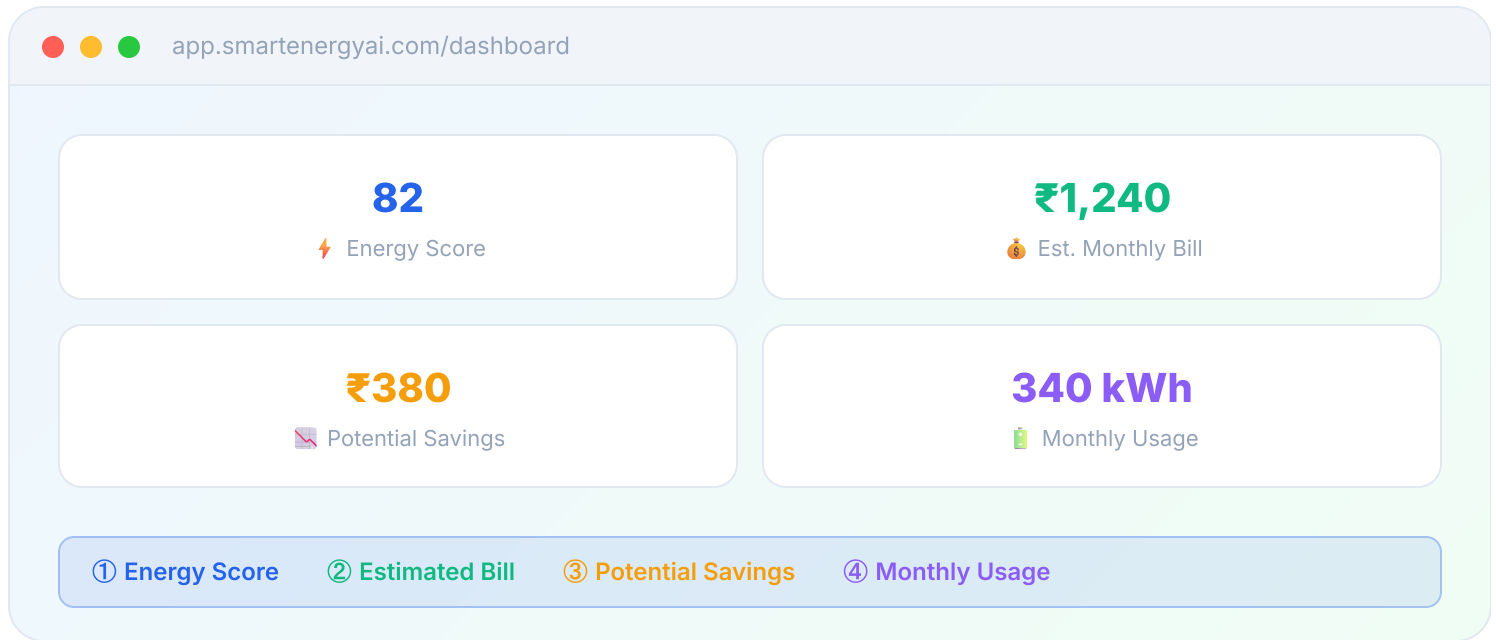
First Results in Minutes

Once you complete Steps 3 and 4, your first personalized energy report is ready within 60 seconds.

 CORE FEATURE

Dashboard Guide

Your central command center for all energy insights, bills, and AI recommendations.



① Energy Score

A 0–100 composite score representing your household's energy efficiency. Above 80 is excellent; below 50 indicates significant optimization opportunities. The score updates each time you add new data or run an audit.

② Estimated Monthly Bill

AI-predicted bill based on your current usage patterns and your DISCOM's slab tariff rates. Updated in real-time as you add or modify appliances.

③ Potential Savings

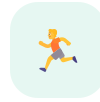
The total monetary savings achievable if you implement the top AI recommendations. Click this card to jump directly to the Recommendations tab.

④ Quick Actions



Add New Bill

Quickly upload a new electricity bill without navigating to the Bill Analyzer.



Run New Audit

Start a fresh energy audit with updated appliance usage data.



Ask AI Assistant

Open the AI chatbot for instant energy queries and personalized advice.



Data Freshness

For most accurate predictions, upload a bill at least once a month. Stale data (older than 60 days) is shown with a  indicator.

 CORE FEATURE

Home Energy Audit

Perform a comprehensive audit of all household appliances to get a precise consumption breakdown and AI-powered report.



Appliance Selection

Choose from 50+ pre-loaded appliances or add custom devices with custom wattage values.



Usage Input

Enter average daily usage hours for each appliance. Weekly and monthly views available.



Budget Setting

Set a monthly bill target. The AI will prioritize recommendations to help you hit it.



AI Simulation

Run real-time simulator models for upgraded LED light combinations or inverter-type appliance configurations.

1

Add Household Appliances

Categorized by Lighting, Cooling, Kitchen, Laundry, Entertainment, and Custom. Click to append appliances to your list.

2

Enter Power Specs & Runtimes

Slide active sliders or write daily usage hours. Enter custom wattages for unusual electronics or older appliances.

3

View instant interactive pie-charts

Visualize consumption by device category (Cooling, Lighting, Cooking) and click to view target savings models.



Lighting upgrades

Dual simulation for LED Bulbs and Tube Lights allows you to see concurrent savings models for upgraded light structures.



Cooling optimization

Setting your AC thermostat to 24°C rather than 18°C saves 6% energy per degree, easily modeled in our simulator.

 CORE FEATURE

Bill Analyzer

Learn how the AI scan extracts metrics from any scanned electricity bill image or PDF.

Multi-layered AI OCR Pipeline

Our platform handles scans with a sequential fallback deck:



Sanity limits & exclusion filters

Extracts amounts strictly between ₹50 and ₹99,999 to avoid capturing customer numbers. Exclusions clean out names matching headers (like "Energy Charges").

 AI INSIGHTS

AI Recommendations

Personalized saving tips curated specifically for your home size, DISCOM tariff, and climate zone.



Lighting Upgrades

Replacing standard tube lights with T5 LEDs yields high ROI, tracked separately by our engine.



Smart Cooling

Learn how shifting AC cooling times reduces bills during peak tariff hours.

 SOLAR ENERGY

Solar ROI Calculator

Simulate net-metered installations and estimate your 25-year compounding return on investment.

6.4 yrs

Estimated Payback
Period

₹4.2L

25-Year Cumulative
Savings

25-Year Balance SVG Projection



 AUDIT HISTORY

My History

View uploaded bills, past audits, and dynamic historical savings profiles.



Historical sync and edit mode

Adjust scanned bill data or override values anytime to correct OCR errors. Database changes cascade and update history charts instantly.

⚙ APP SETTINGS

Settings

Configure interface options, units, DISCOM providers, and alert parameters.

Theme Preference Visual layout style	Choose between Light and Dark mode. Toggle dynamically using keyboard shortcuts.
DISCOM Tariff slab Provider settings	Select your current DISCOM (e.g. APEPDCL) to calibrate the bill analyzer matching algorithm. Custom rates can also be entered.

Profile

Your user details, completion ring, and savings metrics overview.



Rahul Sharma

rahul.sharma@email.com · Hyderabad, Telangana

APEPDCL Provider

Solar Connected

85% Complete

85%

Completion

[? FAQ](#)

Frequently Asked Questions

Find answers to common questions about Smart Household Energy.



How accurate is the AI bill prediction?

Our AI bill prediction achieves an average accuracy of 94-98% when your profile is complete. It factors in historic seasonal adjustments and your DISCOM's specific slab tariff structure.



Can I edit data imported from a scanned bill?

Yes! Navigate to My History, click any entry, and select "Edit Extracted Fields" to override scan results. All calculations refresh immediately.

Troubleshooting

Resolve standard app and scanner errors instantly with this checklist.

 **OCR Scan Failed**

Bill text not extracted correctly

→

 **Solution**

Ensure the upload image is clear (at least 150 DPI) and has sufficient contrast. Avoid uploading crumpled bills or pages with severe shadows.

 Keyboard Shortcuts

SHORTCUT	ACTION	AVAILABILITY
Ctrl + K	Open global search overlay	All sections
Ctrl + D	Toggle Dark / Light visual theme	All sections
Esc	Close open modal or overlay dialogs	All pages

[💬 GET HELP](#)

Contact Support

Get in touch with our team for questions about your profile or tariff database issues.



Email Support

Receive responses from developer support staff within one business day.



Community Forum

Ask questions and discuss efficiency guidelines with other solar adopters.

⚡ Built with ❤️ for sustainable energy

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